DATA ITEM DESCRIPTION
Title: Software Reuse Plan and Analysis

Number: DI-SESS-82042
AMSC Number: N9657
DTIC Applicable: No
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20160524
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2016-024

Use/relationship:

The Software Reuse Plan and Analysis describes a developer's intent to reuse one or more software products, including the software portion of firmware.

The Software Reuse Plan and Analysis provides the acquirer insight into the developer's intent for reusing software products in order to highlight any assumptions, risks, and costs that may be associated with the developer's overall design strategy or methodology.

This Data Item Description contains the format, content, and intended use information for the data product resulting from the work task described by the contract.

Requirements:

1. Referenced documents. This section shall list the number, title, revision, and date of all documents referenced in this document. This section shall also identify the source for all documents not available through normal Government stocking activities.

2. Format. The Software Reuse Plan and Analysis shall adhere to the format described under Content.

3. Content: The Plan shall contain the following:

3.1. Provide a list, in tabular format, of the reuse software products at their lowest level available (CSCI, CSC, CSU, or software to be programmed into a firmware device) to include the size of the software products in source lines of code (SLOC). The list shall also indicate if the reuse software products are intellectual property (IP), government off-the-shelf (GOTS), commercial off-the-shelf (COTS), or open source.

3.2. Detail the lineage of the reuse software products and how they compare or relate to the target system. The detail shall include: (1) the context in which the reuse software products were originally intended, (2) history of the reuse software products from their origin to their current state, (3) portions of the target system to which the reuse software products map, (4) similarities between the reuse software products and the portions of the target system to which they map, and (5) differences between the reuse software products and the portions of the target system to which they map.

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3.3. Detail the software development process/ level of rigor which was applied to the reuse software products.

3.4. Detail the ability of the reuse software products to provide the required capabilities and meet any constraints for the target system.

3.5. Detail the ability of the reuse software products to provide the required safety, security and privacy for the target system.

3.6. Identify the reuse software products that contain more functionality than necessary for the target system.

3.7. Identify the reuse software products that are categorized as safety critical for the target system and whether the reuse software products are accompanied by safety artifacts.

3.8. List of Cyber-Security requirements, architecture, design, integration and test artifacts for the reuse software products showing mitigation of:
   a. Embedded malicious code.
   b. Exploitable vulnerabilities.
   c. Known risks.

3.9. Detail open system characteristics of the reuse software products, including:
   a. Low coupling.
   b. Modularity.
   c. Processor independence.

3.10. Detail reliability/maturity of the reuse software products, as evidenced by established track records.

3.11. Detail the defect history of the reuse software products.

3.12. Detail the testability of the reuse software products, including the test environment.

3.13. Detail the interoperability of the reuse software products with other systems and system-external elements.

3.14. Detail all fielding issues associated with the reuse software products.

3.15. Detail restrictions on copying/distributing the reuse software products themselves or associated artifacts, including documentation.

3.16. Detail license or other fees applicable to the reuse software products.

3.17. Detail maintainability of the reuse software products:
   a. Likelihood the reuse software products will need to be changed.

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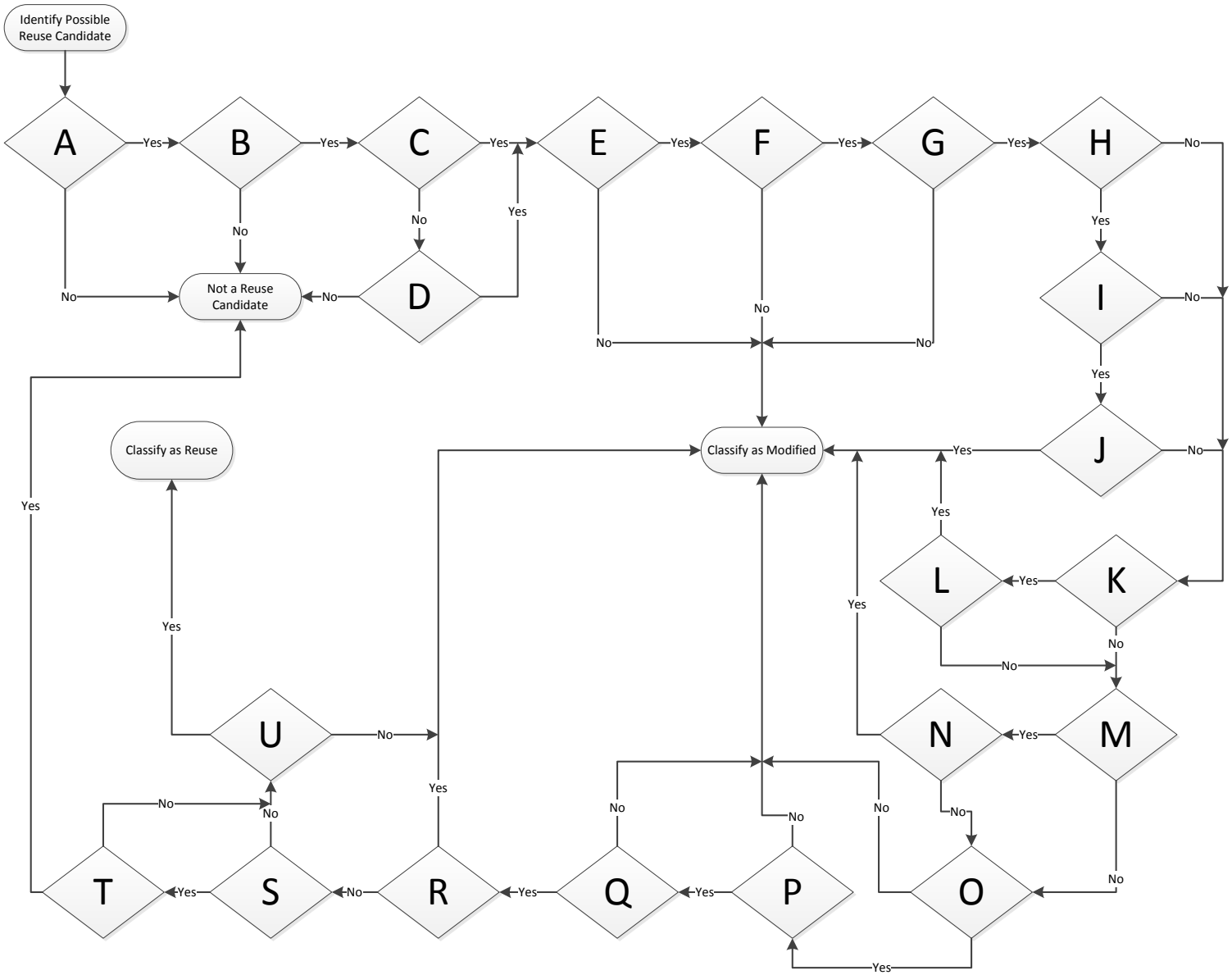
b. Feasibility of accomplishing that change, including the availability of the software development environment.
c. Availability, currency, and quality of existing artifacts associated with the reuse software products, including requirements documents, design documents, interface documents, test plans, test procedures, and source files.
d. Likelihood that the current version of the reuse software products will continue to be supported.
e. Impact on the target system if the current versions of reuse software products are not supported.
f. The acquirer's data rights to the reuse software products and associated artifacts, including requirements documents, design documents, interface documents, test plans, test procedures, and source files.
g. Warranties available for the reuse software products and any associated costs for the warranties.

3.18. Detail the short and long term cost impact of incorporating the reuse software product s into the target system.

3.19. Using the provided flowchart (section 3.20), associated questions (section 3.21) and table example (section 3.22) below, evaluate the reuse software products at their lowest level (CSCI, CSC, CSU, or software to be programmed into a firmware device) and provide the results as part of this deliverable.

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3.20 Flowchart



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3.21 Questions from Flowchart

A. Do the software requirements for the system under development match or form a subset of the software requirements of the reuse code?

B. Reuse code developed to an accepted standard (IEEE-12207, ISO 9000, MIL-STD-498, DOD-STD-2167A)?

C. Reuse code developed to the level of rigor as identified in MIL-STD-882E or Design Assurance Level (DAL) as identified in DO-178B/C to insure sufficient test activities executed during development of the reuse code?

D. Other evidence of sufficient white and black box test rigor?

E. Has the pedigree of the reuse code for mission critical functions, from other systems, been examined for embedded malicious code, exploitable vulnerabilities, and minimizing risk?

F. Was the Common Attack Pattern Enumeration and Classification (CAPEC) used to assist with:

a) Design and architecture considerations
b) Selection of programming language(s)
c) Use of COTS and open source code

G. Was design-based evidence for cybersecurity functions/features documented?

H. Are there features in the reuse code not necessary for the new system?

I. Are there safety implications of inadvertently activating these features?

J. Do the mitigation steps require the removal of code?

K. Are there any cybersecurity implications of inadvertently activating those features?

L. Do the mitigation steps require the removal of code?

M. Are there anti-tamper implications of inadvertently activating those features?

N. Do the mitigation steps require the removal of code?

O. Is the design being reused?

P. Is the software test environment being reused?

Q. Are the test procedures or automation testing scripts being reused?

R. Does this code need to be refactored?

S. Is this reuse to be part of an open system as defined by the NAVAIR Instruction?

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T. Does this reuse candidate contain IP or other restrictive provisions on government usage and/or data rights?

U. Does the documentation (design, models, requirements, etc.) exist and is it available for use for the new program?

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3.22 Sample Table

Function #	CSCI, CSC or CSU Name	DECISION POINTS (Y/N)																					Development Tools Available? (Y/N)	Category (**R**euse, **M**odified, **N**ot for **R**euse)
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U		
1	CSU Name	Y	Y	Y	N																		Y	M
2	CSU Name	Y	N																				Y	NfR
3	CSU Name	Y	Y	Y	Y	Y	Y	Y	N	-	-	Y	N	Y	N	Y	Y	Y	N	N	N	Y	Y	R
4	CSU Name	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y												N	M
5	CSU Name	Y	Y	Y	Y	Y	Y	Y	N	-	-	Y	N	Y	N	Y	Y	Y	N	Y	Y		N	NfR
6	CSU Name	Y	Y	Y	Y	Y	Y	Y	N	-	-	Y	N	Y	N	Y	Y	Y	N	Y	N	Y	Y	R
7	CSU Name																							
8	CSU Name																							
9	CSU Name																							
10	CSU Name																							

End of DI-SESS-82042

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